

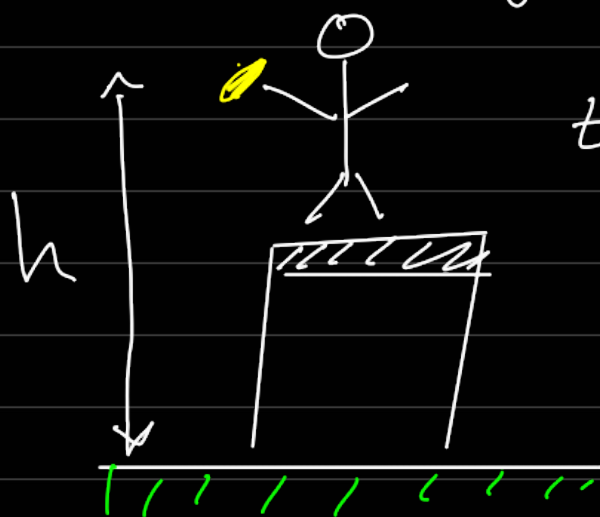
Practice for PS3 Q1:

• Free-falling objects:

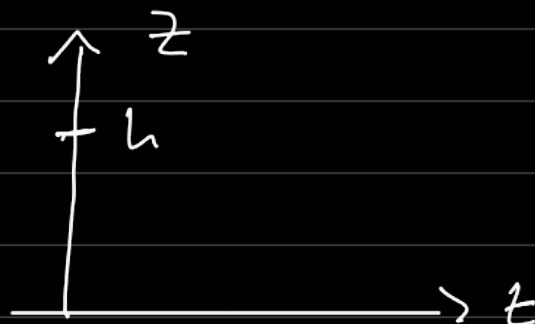
We said $\vec{r}(t) = (x(t), y(t), z(t))$

describes the motion of some object in space $\mathbb{R}^3$. For free fall, the object is only moving along the z -direction

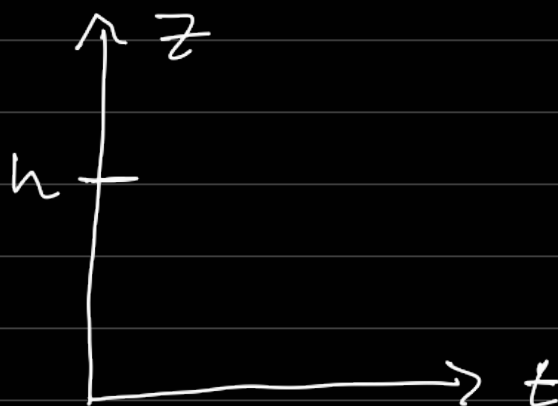
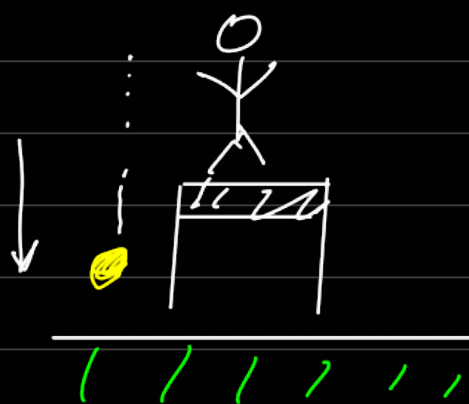
Test



$t = 0$



$t > 0$



- Think about dropping a ball from a roof. If there is no wind then the ball goes straight down.

- Let's model this using what we learned

Newton's Laws:

"(Sum of forces) = m (acceleration)"

in words and in symbols this is

$$\sum_i \vec{F}_i = m\vec{a}$$

- At $t=0$ (see above) we are holding the

ball (●) at a height h . It is at rest as well, so its initial velocity & acceleration are zero.

$$\vec{r}(t) = (0, 0, z(t))$$

- Since it only moves along the z -direction
Recall,
$$F_z = m \frac{d^2 z}{dt^2}$$



Gravity: Near the surface of the earth objects feel a gravitational force which "pulls" them down to the surface.

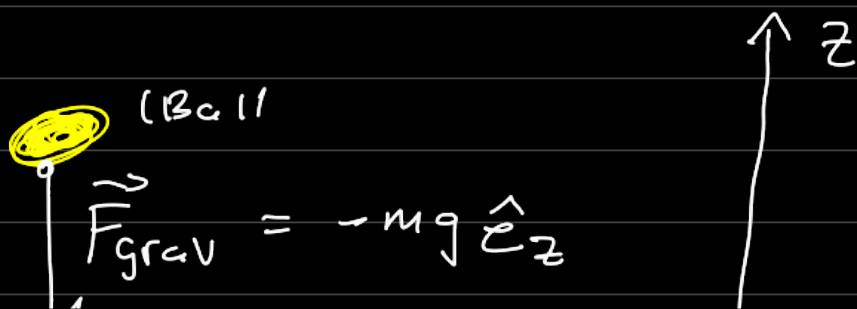
All objects fall with the same constant acceleration:

$$\vec{F}_{\text{grav}} = m \vec{a}_{\text{grav}}$$

We call this acceleration g so we have $a_{\text{grav}} = -g$

$$\boxed{F_{\text{grav}} = -mg}$$

- This is negative because the force vector points down. We draw this as a free diagram



- Under free fall: only the force of gravity is acting on the ball
- Going back to ~~the~~ above:

$$F_z = F_{\text{grav}} = -mg = m \frac{d^2 z(t)}{dt^2}$$

note the mass cancels out:

$$-g = \frac{d^2 z(t)}{dt^2}$$

* g is a constant: $g \approx 9.8 \text{ m/s}^2$

→ So this is simply a second order ODE and we have initial conditions as well

$z(t)$: position of ball along z -axis
 as a function of time
 $h = z(0)$: initial height of ball at time $t=0$
 $z'(0) = v(0)$: initial velocity at time $t=0$
 $-g = \frac{d^2 z}{dt^2}$: we call this the equation of motion as it describes how the ball will move

The time $t=0$ is just the moment right before we drop the ball. Again, we said the ball was at a height h and had zero velocity (its not moving). With this we can solve for $z(t)$:

$$-g = \frac{d^2 z}{dt^2}$$

integrate once:

$$-gt + C_1 = \frac{dz}{dt}$$

integrate again

$$-\frac{1}{2}gt^2 + tC_1 + C_2 = z(t)$$

C_1 : Some constant which we'll find from initial conditions

Another constant "

so we found

$$z(t) = C_2 + tC_1 - \frac{1}{2}gt^2$$

But we know

$$z(0) = h$$

So

$$z(0) = C_2 = h$$

$$\text{so } C_2 = h$$

$$z(t) = h + tC_1 - \frac{1}{2}gt^2$$

To find C_1 we see above that

$$z'(0) = 0 = v(0) \quad (\text{initial velocity})$$

$$z'(0) = C_1 - g \cdot 0 = C_1 = 0$$

$$\text{so } C_1 = 0$$

Thus

$$z(t) = h - \frac{1}{2}gt^2 \quad (\text{Position of Ball})$$

$$\frac{dz}{dt} = v(t) = -gt$$

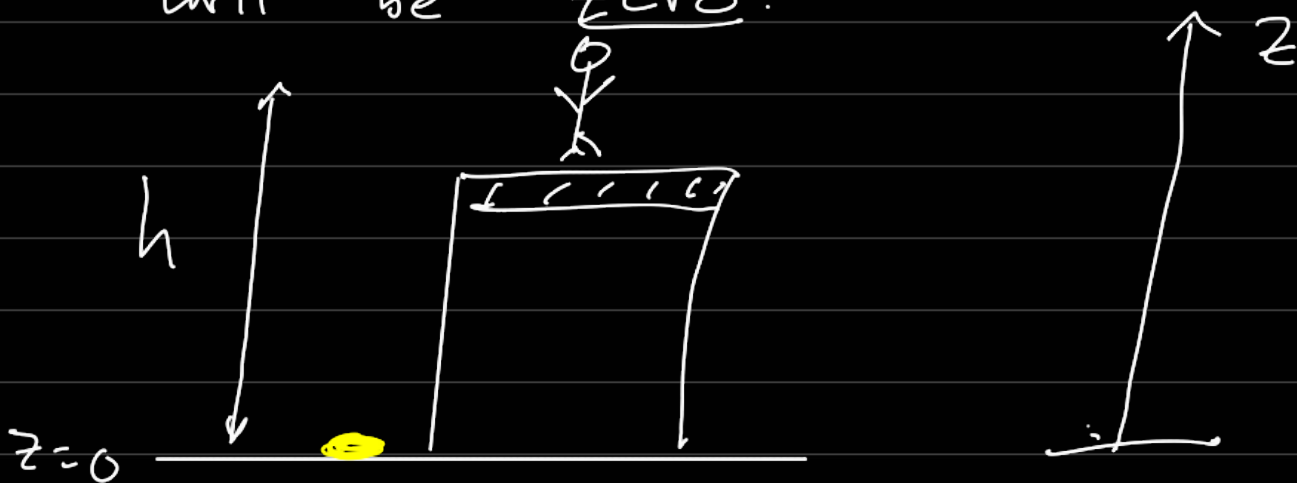
(Velocity of Ball)

$$\frac{d^2 z}{dt^2} = -g$$

(acceleration of Ball)

$$dt^2 \quad \checkmark$$

From the position we know there is some time when the ball will hit the ground. At this time the height will be zero:



T_0 : time for the ball to hit the ground

$$Z(T_0) = 0 = h - \frac{1}{2}gT_0^2$$

Solving for T_0

$$T_0 = \sqrt{\frac{2h}{g}}$$

(time for the ball to reach the ground from falling from a height h)

- Now, what does Q1 mean by scaling? Well, this means we introduce a new way of describing the system:

$$\tilde{z}(t) = \frac{Z(t)}{h} = \frac{1}{h}(h - \frac{1}{2}gt^2)$$

↑
New

So we are rescaling all of our variables

including time

$$\tilde{t} = \frac{t}{T_0}$$

↑
New

so $t = \tilde{t} T_0$, plug this in

$$\tilde{z}(t) = \frac{1}{h} \left(h - \frac{1}{2} g \tilde{t}^2 T_0^2 \right)$$

Plug in $T_0^2 = \frac{2h}{g}$

$$\tilde{z}(t) = \frac{1}{h} \left(h - \frac{1}{2} g \left(\frac{2h}{g} \right) \tilde{t}^2 \right)$$

$$= \frac{1}{h} \left(h - h \tilde{t}^2 \right)$$

$$= \frac{h}{h} (1 - \tilde{t}^2)$$

$$= (1 - \tilde{t}^2)$$

so $\tilde{z}(t) = \tilde{z}(\tilde{t}) = 1 - \tilde{t}^2$

↑

and so new

$$\tilde{z}(\tilde{t}) = 1 - \tilde{t}^2$$

New Equation of motion in our scaled terms. What's the point to this? Well, you'll note we've gotten rid of the "information" about the problem and we can work with how the system

evolves without having to worry about the values of g, h, T_0 :

$$\begin{aligned}\tilde{z}(\tilde{t}) &= 1 - \tilde{t}^2 && \text{(Position)} \\ \tilde{z}'(\tilde{t}) &= \tilde{v}(\tilde{t}) = -2\tilde{t} && \text{(Velocity)} \\ \tilde{z}''(\tilde{t}) &= \tilde{a}(\tilde{t}) = -2 && \text{(acceleration)}\end{aligned}$$

$$\tilde{z}(0) = 1 \quad \text{and} \quad \tilde{z}'(0) = 0$$

as asked.

The kinetic Energy in the base units is

$$\begin{aligned}T_{SI} &= \frac{1}{2} m v^2 = \frac{1}{2} m (z')^2 = \frac{1}{2} m (-gt)^2 \\ &= \frac{1}{2} m g^2 t^2 \\ \text{replace } t &= \tilde{t} T_0\end{aligned}$$

$$T_{SI} = \frac{1}{2} m g^2 (\tilde{t} T_0)^2 = \frac{1}{2} m g^2 T_0^2 \tilde{t}^2$$

$$T_0^2 = \frac{2h}{g}, \text{ so}$$

$$\begin{aligned}T_{SI} &= \frac{1}{2} m g^2 \left(\frac{2h}{g} \right) \tilde{t}^2 \\ &= m g h \tilde{t}^2\end{aligned}$$

we're almost there, we know that

$$\tilde{z}'(\tilde{t}) = -2\tilde{t}$$

$$\text{so } \tilde{t}^2 = \left(\frac{2\tilde{t}}{2} \right)^2 = \frac{1}{4} (-2\tilde{t})^2 = \frac{1}{4} (\tilde{z}')^2$$

Plugging this in:

$$T_{SI} = m g h \left(\frac{1}{4} (\tilde{z}')^2 \right)$$

But the problem states:

$$T_{kin} = \frac{1}{4}(\dot{z}')^2, \text{ so}$$

$$T_{SI} = mgh \quad T_{kin}$$

dividing by mgh :

$$T_{kin} = \frac{T_{SI}}{mgh} = \frac{T_{SI}}{E_0}$$

So $E_0 = mgh$

↳ This is the gravitational Potential energy.

② Turns out E_0 is exactly the potential with

$$V(z) = mgz$$

and

$$V(z=h) = mgh = E_0$$

or in our new variables

$$\tilde{V}(\tilde{z}) = \tilde{z}$$

$$-\frac{d\tilde{V}(\tilde{z})}{d\tilde{z}} = -1$$

and so for $F_g(z) = -1$

$$F_g(z) = - \frac{\partial V_{eff}}{\partial z}$$

③ we did this above.

